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Lathe operations -

The working principle of lathe machine, job rotates and cutting tool is stationary the various operation that can be perform on a lathe machine are turning, boring, drilling, reaming, parting, etc.

(4) Turning operation :-

NOTE - It involves reducing the diameters of a rotating work piece.

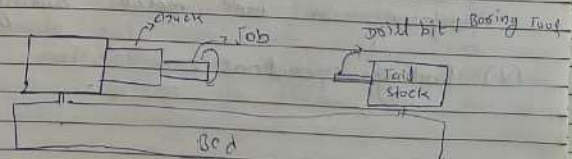
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(ii) Facing operations -

(iii) knowing operation -

(IV) Drilling & Boring :-



Quick Return mechanism:-

It is a mechanism on which most of the shaper machine utilize crank and slotted lever to convert rotary motion into reciprocating motion. It consist of rotating crank, slider, slotted lever, and connecting rod.

(i) Crank :-

It is driven by a motor rotates at a constant speed.

(ii) Slider :-

It is connected to the crank it slides within the slotted lever.

(iii) Slotted lever :-

It oscillate as the slider moves.

(iv) connecting rod :-

It connects slotted lever to the Ram.

(v) Cutting stroke :-

The crank rotates slider moves within the slotted lever and slotted lever oscillates as a result of which the Ram and Tool move forward. It is known as cutting stroke.

Subject

Return stroke :-

As the grind continues to rotate the slider moves again because of slotted lever the return stroke completed faster than cutting stroke. As the known productive time reduce the productivity of machine is increased.

Shaper operations :-

(1) Horizontal cutting :-

It is a process in which the horizontal surface of an object is to be machined by shifting the workpiece in the opposite direction of the cutting tool. This process is generally utilized to cutting grooves.

(2) Inclined cutting operation :-

In this process all the cuts are created at an angle other than 90° to the horizontal and vertical plane after moving the workpiece on the table as per desired cutting angle.

(3) Irregular cutting operation :-

In this process the tool utilized with a circular nose and

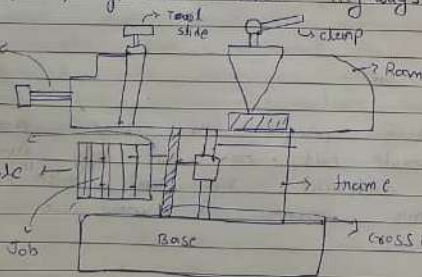
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The helical grooves

per the thread requirement

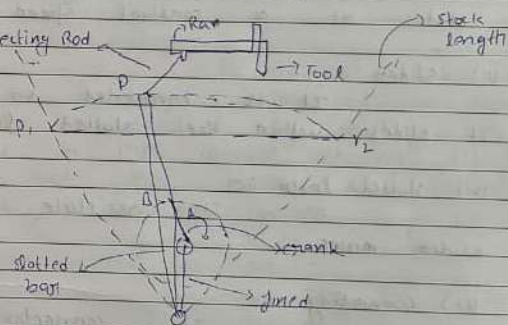
slot, grooves and key ways



Date: 9/8/25

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Subject

Date: 18/8/25
MON TUE WED THU FRI SAT SUN

It is a operation in which cylindrical holes are generated into solid material. The work piece is rotating and the axis of hole is maintain and with the help of drill bits the drilling operations is to be completed.

Boring :- It is defined as enlarging a pre-drilled hole is to be form with the help of boring tool.

NOTE :- (i) For drilling operations boring is not essential but for boring operations the drilling is to be essential.

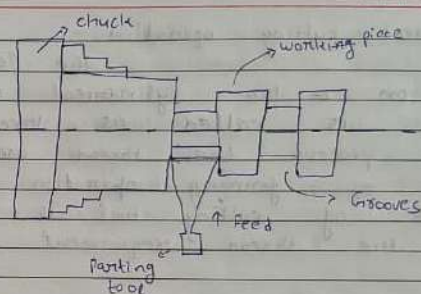
(ii) As the drill bits of each and every dimension is not available in the market but sometimes as per customer requirements the different sizes of internal hole is to form to obtain this the boring operation is to perform in the metal cutting industries.

(v) Parting Operations :-

Feed $\rightarrow$ Revol

Subject

Date: / /
MON TUE WED THU FRI SAT SUN



NOTE :-

Feed :- It is define as the distance cutting tool advance into or along the work piece with each revolution of work piece.

Unit $\rightarrow$ mm/Revolution

It is an

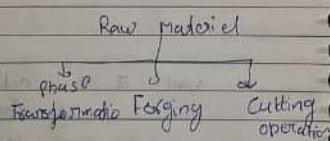
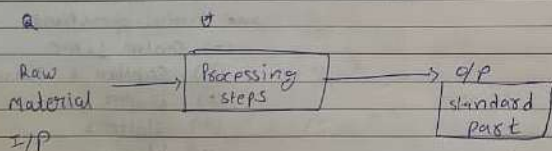
In this process the standard product is separate out or cut off from parent material. The tool which is use in this process is know as parting tool.

Subject

Unit-III → TIG, MIG, Resistance welding, EDM, LBM, Thermit welding

Unit-IV → special purpose machine tools
 > Automates, Semi Automates, Tool layout
 > sheet metal working
 > Compound die, progressive die, combination dies

Unit-V → CNC Machines
 > NC Machines → open and close loop machine
 > DNC / DNC / NC
 > Adaptive control
 > MCV



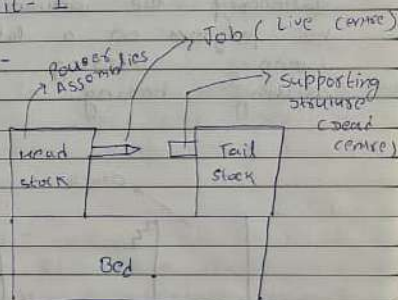
Forging → heat the material up to red hot

Subject

stage apply external forces and obtain the final product.

Unit-I

Centre lathe :-



Universal machine ← Lathe

Introduction to Lathe -

It is a machine tool in which the raw material is processed with the help of single point cutting tool to obtain a standard product as per customer requirement. The basic principle of this machine cutting tool is stationary job rotates about its axis.



Subject

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Shaper

Planer

1. The workpiece is stationary and the cutting tool moves on the ram move in backward & forward direction across the workpiece.

The tool is stationary. Workpiece travels back and forth under the tool.

2. Shaper always use for shaping smaller jobs.

When the job size is large as 6 meters wide and twice as long can be machine easily on planer.

3. It is a light duty machine.

It is a heavy duty machine.

4. It can only use one cutting tool at a time.

It can use several tool simultaneously can be use on planer.

5. The working mechanism is quick return mechanism.

where as planer they are working on gear arrangements and hydraulic arrangements.

6. It is less rigid and less robust.

as compared to shaper it is more rigid and give more accuracy on the machine surface.

Subject

Slotter :-

It is a machine which is considered as reciprocating type of machine generally operated as vertical shaper. It is always use for specific operations.

(i) It is use for shaping internal and external blind holes.

(ii) It is use for machine dies and punches.

(iii) It is use for machining vertical surfaces.

(iv) also use to machine angular or incline surface.

(v) It also helps to machine circular or semi-circular convex and concave surfaces.

In this the tool job is in stationary position and tool moves on the ram in up and down direction. It is always use to create slots when the geometry of an object is very complicated. It is also consider as light duty machine.

Applications of Lathe :-

- (1) It is generally use for machining engine component also use for wooden product and wood working industries. For producing precision part in various metal working industries it also helps to repair critical components like pumps valve in oil and gas industries. It also have produce and repair unique components which are use in aircraft and automobile industries.

Applications of shaper :-

- Shaper machine helps to produce dies and punches. It also helps to smooth rough surface on metal and other materials. It also helps to machine irregular and complicated profile job.

Applications of planer :-

- Planer are generally use to flat and shape large and heavy metal parts.
- It also helps to produce parallel flat surfaces with high precision on large size plates.

- It also helps to cut various slots and grooves on metal surface.

Application of slotter :-

- They are generally use to cut internal and external grooves in various components with high precision.
- It is also use to cut keyways in shaft.
- It also helps to repair and maintain various types of components in machinery.
- These types of machines are generally used in rolling mills, paper mills and various shape building industries.

Date: 08/09/25
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Subject

① Centre Lathe Capstan and Turret lathe

- 1 It is a manually operated lathe It is a semi automatic lathe
- 2 It has only one tool post. So tool changing time is more. Front and rear tool posts are available. Tool changing time is less
- 3 It has tail stock It has turret head instead of tail stock
- 4 Only one tool can be fitted in the tail stock. Sim different tool can be fitted in the turret head
- 5 Number of speeds is less. Number of speeds is more
- 6 Tool changing time is more. Tool changing time is less
- 7 The machine should be stopped for changing tool. Tool can be changed without stopping the machine
- 8 It is not suitable for mass production. It is suitable for mass production.
- 9 No feed stops to control the tool. The tools are controlled by feed stops
- 10 The tool is controlled manually after changing the tool. The tool is controlled automatically.

Date: / /
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Subject

1. Only one operation is done at a time. More than one operation can be done at a time.

② Capstan lathe Turret lathe

- 1 It is a light duty machine. It is a heavy duty machine.
- 2 The turret head is mounted on the ram and the ram is mounted on the saddle. The turret head is directly mounted on the saddle and the saddle slides over the bed ways.
- 3 The saddle will not be moved during machining. The saddle is moved along with the turret head during machining.
- 4 The lengthwise movement of turret is less. The lengthwise movement of turret is more.
- 5 Short work piece only can be machined. Long work pieces can be machined.
- 6 It is easy to move the turret head as it slides over the ram. It is difficult to move the turret head along with saddle.
- 7 The turret head can not be moved crosswise. The turret head can be moved crosswise in some turret lathes.

Subject: _____

8 As the construction of lathe is not rigid, heavy cut can be given.

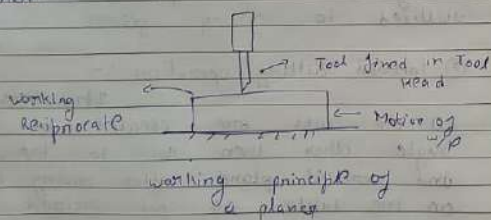
9 It is used for machining work. It is used for machining pieces up to 60mm diameter.

generally it is used to make curve on the surface of the object

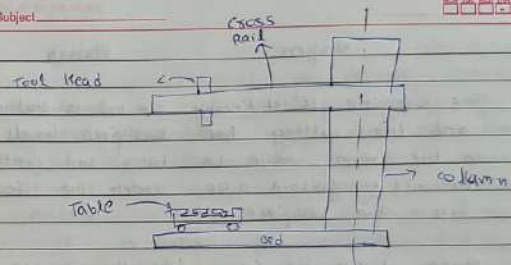
Application of shaper machines:-

They are generally use to create flat surface also use to smooth the rough surface. It is also use to create longways, splines and various internal features. It is also use produce concave and convex shape on the various objects.

Planer:-



Subject



open side planer

When the diameter and length of an object is very large and it is not possible to machine out on shaper we move the job on the machine name planer. It is also use to produce horizontal, vertical or inclined flat surface with help of single piece cutting tool. It is always use for machining large and heavy workpieces. and the working principle of planer - cutting tool is stationary and workpiece which is supported on the table reciprocates. It means tool is fixed and workpiece is in motion.

Q. Differentiate between shaper and planer?